

Applications of differentiation 2

1. Consider a Space-battleship flying along the path $f : [0, 3] \rightarrow \mathbb{R}$, $f(x) = \sqrt{x}e^{-x}$. There is an enemy space station at the point $(0, 1)$. The battleship crew would like to shoot the enemy station from as close as possible. Find the coordinates at which the battleship should shoot from.
 1. As a response, the enemy station launches a drone in a straight line to intercept the battleship on their path. The launch is angled precisely so that the path of the drone is tangential to the battleship's path. Find the possible coordinate(s) at which the battleship can be intercepted by the enemy drone.
 2. Hence, find possible equation(s) describing the enemy drone's path with appropriate domain(s).
 3. Suppose the path of the drone launched by the enemy followed the line $y = ax$. For what values of a will there be no interception of the battleship?
2. The outbreak of a certain virus in a town can be modeled by a function $C(t) = \frac{1000}{1 + e^{-\frac{1}{5}(t-10)}}$, denoting the number of confirmed virus cases, t days after the outbreak.
 1. When does the virus outbreak grow the fastest?
 2. It is recommended to practice social distancing when the rate of increase is above 30. Find the interval at which social distancing should be practiced.
 3. Suppose on day 18, 13 travellers arrived at the town of the outbreak, and it was found that they had already contracted the virus elsewhere. Update the function $C(t)$ to reflect the true number of confirmed cases in the town. (Assume the spreading rate is unchanged by the arrival of new cases)
3. Consider a curve $h(x) = \left(\ln(x) - \frac{3}{2}x^2 - \frac{1}{4}\right)$.
 1. Find the maximal domain and range of h
 2. Find the x intercepts of h
 3. Consider two lines, each tangential to h at an x intercept. Find the equations of each of the lines.
 4. Find the obtuse angle formed by the lines in the form $\tan(\alpha)^{-1} + \tan(\beta)^{-1}$